

data_management_plan

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Data Management Plan

1 Data Collection

The data used in this project are obtained from the publicly available **Palmer Penguins dataset** provided through the **palmerpenguins** R package. The dataset contains morphological measurements of penguins collected at Palmer Station, Antarctica.

For this project, the raw dataset was exported and stored in the project directory under:

data/data_raw/

The processed dataset was created by selecting relevant variables and removing observations with missing values. The cleaned dataset is stored in:

data/data_processed/

No new data were collected for this study.

2 Documentation and Metadata

Documentation for this project is provided through multiple files to ensure transparency and reproducibility.

Project documentation includes:

- **README.md** in the root directory describing the project structure and objectives.
- **DATA_DICTIONARY.md** files within both the raw and processed data folders describing the variables contained in each dataset.
- An **R Markdown manuscript** documenting the analysis workflow.

These files provide metadata describing variable definitions, data processing steps, and analysis procedures.

3 Storage, Backup, Data Security and Access Management

Project files are stored locally within an RStudio project directory and organized into structured folders including:

- data/
- docs/
- project documentation files

Version control is managed through **GitHub**, which tracks all changes to project files and provides a remote backup of the working directory.

An additional copy of the project files is stored on the **Open Science Framework (OSF)**, providing an external backup and facilitating project sharing.

Package management is handled using the **renv** package, which ensures that all required R packages and versions can be restored for reproducibility.

The dataset used in this project is publicly available and contains no sensitive or confidential information.

4 Deposit, Reuse, and Preservation

All project materials including documentation, analysis code, and processed datasets are preserved through the OSF project repository.

The GitHub repository provides version-controlled access to the project files and ensures that previous versions of the analysis workflow are preserved.

The Palmer Penguins dataset itself is publicly available through the **palmerpenguins** R package and can be reused by other researchers following the package documentation.